Output:

Enter plain text : Hello world

Enter shift : 3

Cipher text : KHOOR ZRUOG

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Exp 1: Caesar Cipher

Aim: To implement the Caeser cipher in C by replacing each alphabetic character with the character located a fixed no of positions.

98/100

Algorithm:

-> Read the plaintext and shift value k

-> for each alphabetic character, converting it using

- $C = (P+K) \bmod 26$

-> Preserve the upper case and lowercase letters.

C Program

```
for (i=0; text[i] != '\0'; i++){
    if (text[i] >= 'A' && text[i] <= 'Z')
else if (text[i] >= 'a' && text[i] <= 'z')
    }
Printf ("Chipher text: %s", text);
```

Result:

Thus, the plaintext was successfully encrypted using the Caesar cipher.

Output:

Enter plaintext : Hello world

Cipher text : ITSSG VG KSR

Exp 2 : Monoalphabetic substitution cipher

Aim : To implement a monoalphabetic substitution cipher in which every plaintext letter is mapped to one unique cipher text.

Algorithm :

- create a 26 - character substitutional key
- for every alphabetic character, find its position in the normal alphabetic.
- Replace it with corresponding character from the key.

C Program

```
for (i=0; text[i] != '\0'; i++) {
if (isalpha ((unsigned char) text[i])) {
        P= toupper ((unassigned char) text[i]-'A';
            text (i] = key [P];
    }
}
```

Result: Thus, the plaintext was successfully encrypted using a monoalphabetic substitution key.

Output:

Enter keyword: Monarchy

Enter plaintext: Instrements

cipher text: GAT LM IZ CLRQTX

Exp 3 : Playfair Cipher

Aim : To implement the playfair cipher using a 5x5 matrix constructed from a keyword and encrypt plaintext in pairs of letters.

Algorithm:

- for letters in the same row, replace each other with the letter to its right
- for letters in the same column, replace each with the letter below it
- otherwise, form a rectangle and replace each letter.

Code:

```
for (i=0; key[i]; i++) {
    x = toupper((unsigned char) key[i]) - 'A';
    if (x >= 0 && x < 26 && !used[x]) {
        key_mat[r][c++] = 'A' + x; used[x] = 1;
        if (c == 5) { c = 0; r++; }
    }
}
```

Result: Thus, the plain text was successfully encrypted using the playfair cipher.

Exp 4: HILL Cipher

Aim: To implement the hill cipher using a 2x2 key matrix for block encryption

Algorithm:

-> Divide the plaintext into blocks for two letters

-> Multiply each plaintext vector by the key matrix

-> Reduce every result modulo 26.

Code:

```
for (i=0; i<n; i+=2){
    x = text[i]-'A'; y= test[i+1]-'A'
    out[i] = 'A' + (k[0][0] * x+k[0][1] * y)%26;
    oot[i+1] = 'A' + (k[1][0] * x+k[1][1] * y)%26;
    }
```

Result:

Thus the plain text was successfully encrypted using Hill cipher.

Output:

Enter plaintext : SEND MORE MONEY

cipher text : BENC VXBX KX WJS

Exp 5 : Vigenere cipher

Aim :

To implement a one-time Pad style vigenere encryption using a supplied numerical key stream

Algorithm :

- Convert each plaintext to a value from 0 to 25
- Add the corresponding key value
- Reduce the result Modulo 26.

Code:

```
for (i=0 ; text[i]; i++) {
    if (isalpha ((unsigned char) text[i])) {
        out[i] = 'A' + (toupper ((unsigned char)
                                 text[i]) - 'A'
        j++;
    }
```

Result:

Thus the plaintext was successfully encrypted using the supplied one-time-pad key stream.

output:

Enter plaintext: WEARE DISCOVERED FLEE ATONCE

Enter No. of rails: 3

cipher text: WECRLTEE RDSOEEFEOCAIVDEN

Exp 6: Rail fence transposition

Aim:

To write and execute a C program to implement the Rail fence transposition cipher for encrypting a plain text message.

Algorithm:

- Create a matrix to store the characters in a zig zag pattern
- Place the plaintext characters diagonally across the rails.
- Read the matrix row by row to obtain the cipher text.

Code:

```
if (key < 1 || key > 10) {
        Print ("Invalid number of rails\n");
        return 0;
      }
for (i= 0; i< len; i++) {
        rail [row] [i] = text[i];
        if (row == 0)
          dir= 1
        else if (row == key - 1)
          dir = -1;
      }
```

result:

Thus the C Program for the Rail fence transposition cipher was successfully executed.

output:

Enter plaintext: ATTACK DAWN

No of Columns: 4

Enter key order: 3 1 4 2

cipher text: TKAATNACDTAW

EXP 7: Columnar transposition

Aim:

To write and execute a C Program to implement the Columnar transposition cipher for encrypting a plaintext message using a Numeric key.

Algorithm:

- Arrange the plaintext row by row in a matrix
- fill any remaining empty positions with the character X.
- Read the Columns in ascending order of their key values.

Code:

```
for (i=0; i<cols; i++)
    scanf ("%d", &key[i]);
    rows = (len+ col-1] / cols;
for (i=0; i< rows; i++) {
  for (j=0; j < cols; j++) {
      if (pos < len)
          matrix [i][j] = text [pos++];
      else
          matrix [i][j] = 'x';
  }
```

Result:

Thus the C Program for the Columnar transposition cipher was successfully executed.